A THIRD-DERIVATIVE COUNTEREXAMPLE FOR CHROMATIC POLYNOMIALS

Abstract. Let $P(G,x)$ be the chromatic polynomial of an n-vertex graph G. For $x<0$, set $\mathcal{L}_G(x)=\log\big((-1)^nP(G,x)\big)$. Conjecture 2 of Dong et al. asserts that $\mathcal{L}_G^{(k)}(x)<0$ for every graph G, every $k\geq 2$, and every $x<0$. We refute it by the simple planar generalized theta graph $H=\Theta(8,8,9,9)$: an exact calculation gives

$$\mathcal{L}_H^{(3)}\big(-\tfrac{1}{2}\big)=\frac{5255579329573016681520193060821596608}{18016243181285229121994454910138997625}>0.$$

The same witness violates the upper inequality in their stronger Conjecture 3.

1. Introduction

Let $P(G,x)$ denote the chromatic polynomial of a graph G of order n. Its coefficients alternate in sign, so $(-1)^nP(G,x)>0$ for $x<0$ and

$$\mathcal{L}_G(x):=\log\big((-1)^nP(G,x)\big)$$

is well defined on $(-\infty,0)$. Dong, Ge, Gong, Ning, Ouyang, and Tay proposed the following conjecture [2, Conjecture 2]:

$$\mathcal{L}_G^{(k)}(x)<0\qquad(k\geq 2,\ x<0)\tag{1}$$

for every graph G. They further conjectured that, whenever G is a non-complete graph of order n and Q is a chordal proper spanning subgraph of G,

$$\mathcal{L}_Q^{(k)}(x)<\mathcal{L}_G^{(k)}(x)<\mathcal{L}_{K_n}^{(k)}(x)\qquad(k\geq 2,\ x<0)\tag{2}$$

[2, Conjecture 3]. Ning and Yang proved (1) in the range $x\leq -3.01\Delta(G)k$, where $\Delta(G)$ is the maximum degree [3, Theorem 1.5].

For positive integers $s_1,\dots,s_p$, let $\Theta(s_1,\dots,s_p)$ be the graph obtained from two terminals by joining them with p internally vertex-disjoint paths of lengths $s_1,\dots,s_p$, measured in edges.

Theorem 1. *Let $H=\Theta(8,8,9,9)$. Then H is a simple planar graph on 32 vertices and*

$$\mathcal{L}_H^{(3)}\Big(-\frac{1}{2}\Big)=\frac{5255579329573016681520193060821596608}{18016243181285229121994454910138997625}>0.$$

Consequently, Conjecture 2 of Dong et al. is false, even for simple planar graphs.

2020 *Mathematics Subject Classification.* 05C31, 05C15.

Key words and phrases. chromatic polynomial, logarithmic derivative, generalized theta graph, counterexample.

The witness does not conflict with the positive result quoted above. Here $\Delta(H) = 4$ and $k = 3$, so [3, Theorem 1.5] asserts (1) for H and $k = 3$ only in the range $x \le -3.01 \cdot 4 \cdot 3 = -36.12$, and the witness point $x = -1/2$ lies far outside it. Theorem 1 therefore locates the failure of (1) strictly between -36.12 and 0, where nothing was proved.

Status. What is refuted is (1) as a universally quantified statement over all graphs, all $k \ge 2$ and all $x < 0$; one exact witness settles that. What is not touched: the case $k = 2$ of (1), for which the present witness gives a negative value (see the Remark below); the range $x \le -3.01\Delta(G)k$, which is a theorem; and the lower inequality of (2), since every spanning tree Q of H has $\mathcal{L}_Q^{(3)}(-1/2) = -928/27 < \mathcal{L}_H^{(3)}(-1/2)$. Only the upper inequality of (2) is refuted here, and only at $k = 3$, $x = -1/2$.

Statement numbering. The labels "Conjecture 2" and "Conjecture 3" are those of the accepted version arXiv:1803.08658v3 of [2], which is the only full text we could reach; the printed journal text was not accessible to us and its numbering may differ. Both statements are reproduced in full in (1) and (2) above, so what is refuted here does not depend on the labelling.

2. Exact calculation

For a path of length s with prescribed endpoint colors, set

$$A_s(q) = \frac{(q-1)^s + (q-1)(-1)^s}{q}, \qquad D_s(q) = \frac{(q-1)^s - (-1)^s}{q}.$$

The numerators vanish at $q = 0$, so $A_s, D_s \in \mathbb{Z}[q]$.

The next formula is not new. After clearing the factors of q in the A_{s_i} and D_{s_i} it is algebraically identical to [1, eq. (2.6a)], and for $p = 3$ the authors of [1] attribute it to Read and Tutte. We record a proof only to fix the normalisation used in Theorem 1.

Lemma 2 (Brown, Hickman, Sokal, and Wagner [1])**.** *For positive integers $s_1, \ldots, s_p$,*

$$P(\Theta(s_1, \ldots, s_p), q) = q \left(\prod_{i=1}^{p} A_{s_i}(q) + (q-1) \prod_{i=1}^{p} D_{s_i}(q) \right). \tag{3}$$

Proof. Fix an integer $q \ge 2$, let J be the all-ones $q \times q$ matrix and I the identity. Then $J - I$ is the adjacency matrix of K_q, so the (c, c') entry of $(J - I)^s$ is the number of proper colorings of the internal vertices of a path of length s whose endpoints are colored c and c'. Since $J^2 = qJ$, induction on s gives

$$(J - I)^s = D_s(q)\, J + (-1)^s I$$

(equivalently: $J-I$ has eigenvalues $q-1$ once and -1 with multiplicity $q-1$). Its diagonal entries are $D_s(q)+(-1)^s = A_s(q)$ and its off-diagonal entries are $D_s(q)$, so that count is $A_s(q)$ when $c = c'$ and $D_s(q)$ when $c \ne c'$. Now color one terminal of $\Theta(s_1, \ldots, s_p)$, in q ways. The other terminal either receives the same color, the p paths then contributing $\prod_i A_{s_i}(q)$, or one of the other $q-1$ colors, contributing $(q-1)\prod_i D_{s_i}(q)$. This gives (3) for every integer $q \geq 2$, and since both sides are polynomials in q, the identity follows. $\square$

Proof of Theorem 1. The vertex count is

$$2+(8-1)+(8-1)+(9-1)+(9-1)=32.$$

All path lengths are at least two, so H is simple, and the generalized theta construction is planar. Since $|V(H)|$ is even, $\mathcal{L}_H(x)=\log P(H,x)$ for $x<0$.

Applying (3) to H and differentiating at $q=-1/2$ gives

$$P\left(H,-\tfrac{1}{2}\right)=\frac{16528743341798025}{4294967296}, \qquad P'\left(H,-\tfrac{1}{2}\right)=-\frac{38758016767164565}{536870912},$$
$$P''\left(H,-\tfrac{1}{2}\right)=\frac{5549131014505347}{4194304}, \qquad P'''\left(H,-\tfrac{1}{2}\right)=-\frac{1586794036505230809}{67108864}.$$

For any nonvanishing function f,

$$(\log f)'''=\frac{f'''}{f}-3\frac{f'f''}{f^2}+2\left(\frac{f'}{f}\right)^3.$$

Substitution yields

$$\mathcal{L}_H^{(3)}\left(-\tfrac{1}{2}\right)=\frac{5255579329573016681520193060821596608}{1801624318128522912199445491013899762 5} > 0,$$

which contradicts (1). $\square$

Remark. The failure is narrow, and a reader sampling other values will see negative numbers. At the same point the second derivative is still negative,

$$\mathcal{L}_H^{(2)}\left(-\tfrac{1}{2}\right)=-\frac{5580257286440852356107116 8}{6872416789892908604843025}=-8.1197\ldots,$$

so the sign change is a third-order phenomenon; and on the grid of step 10^{-3} in $[-3/5,-3/10]$ the inequality $\mathcal{L}_H^{(3)}(x)>0$ holds exactly at the points with $-0.518 \leq x \leq -0.433$, the sign being negative at both neighboring grid points. In particular $\mathcal{L}_H^{(3)}(-1)<0$ and $\mathcal{L}_H^{(3)}(-1/4)<0$: the witness is specific to $k=3$ and to x near $-1/2$.

Corollary 3. *Conjecture 3 of Dong et al.* [2, Conjecture 3] *is false. More precisely, its upper inequality fails for $G=H$, $k=3$, and $x=-1/2$.*

Proof. For $x<0$,

$$(-1)^n P(K_n,x)=\prod_{j=0}^{n-1}(j-x), \qquad \mathcal{L}_{K_n}^{(k)}(x)=-(k-1)!\sum_{j=0}^{n-1}(j-x)^{-k}<0.$$

Theorem 1 gives $\mathcal{L}_H^{(3)}(-1/2)>0$, so the upper inequality in (2) fails. The graph H is connected and noncomplete, and any spanning tree of H is a chordal proper spanning subgraph, so all hypotheses of Conjecture 3 are satisfied. $\square$

3. Reproduction and verification

Reproduction. Lemma 2 determines $P(H,q)$ from the parameters $(8,8,9,9)$ alone, so the calculation can be redone in a few lines of exact rational arithmetic: expand (3) with $p=4$ and $(s_i)=(8,8,9,9)$, take the first three derivatives at $q=-1/2$, and substitute into the identity for $(\log f)'''$ above. For a concrete labelling, take the terminals to be 0 and 1 and number the internal vertices consecutively along the four paths,

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0-2-3-4-5-6-7-8-1,      0-9-10-11-12-13-14-15-1,
0-16-17-18-19-20-21-22-23-1,      0-24-25-26-27-28-29-30-31-1,
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which are the 34 edges of H; the corresponding graph6 string is the concatenation of the two lines

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_PCGGD@_??_@?@??_?I?@_????G??G??C??@???G???g??@_???????_???G???@
????C????G????I????C
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Exact verification. The computation reported here was re-run independently, on a separate machine and from the graph6 string printed above, in exact integer and rational arithmetic throughout. It decodes and re-encodes H; confirms by an explicit bijection that the decoded graph is $\Theta(8,8,9,9)$ with 32 vertices, 34 edges and $\Delta(H)=4$; checks planarity twice, by a Kuratowski branch-vertex count and by tracing the faces of an explicit rotation system against Euler's formula; computes $P(H,q)$ both from (3) and, independently, by a generic proper-coloring count at $q=0,\dots,32$ followed by exact interpolation; recomputes the four derivatives displayed above and $\mathcal{L}_H^{(3)}(-1/2)$ by two routes; and rechecks the hypotheses of (2), including that all 2448 spanning trees of H are chordal proper spanning subgraphs. All 80 of its checks passed, and every value it recomputed agreed with the value printed here. The program used for that re-run, together with its transcript, accompanies this note; it is not needed in order to redo the calculation, since every input it consumed is printed above.

One limit of that re-run should be recorded. It also ran a census of generalized theta graphs, which is not among the claims of this note and which the re-run itself records as bounded: the census is complete only for order at most 34, larger orders were not enumerated, and no graph outside the generalized theta family was examined. Nothing here, therefore, asserts that H is a counterexample of least order, either among generalized theta graphs or among graphs.

References

[1] J. I. Brown, C. A. Hickman, A. D. Sokal, and D. G. Wagner, *On the chromatic roots of generalized theta graphs*, J. Combin. Theory Ser. B **83** (2001), no. 2, 272–297. doi:10.1006/jctb.2001.2057; arXiv:math/0012033.

[2] F. M. Dong, J. Ge, H. L. Gong, B. Ning, Z. Ouyang, and E. G. Tay, *Proving a conjecture on chromatic polynomials by counting the number of acyclic orientations*, J. Graph Theory **96** (2021), no. 3, 343–360. doi:10.1002/jgt.22617. Statement numbers

are quoted from the accepted version arXiv:1803.08658v3, the only full text we could access.

[3] B. Ning and Y. Yang, *On derivatives and higher-order derivatives of chromatic polynomials*, arXiv:2604.13221v2, 2026. arXiv:2604.13221.